

Software Project Management Quiz

Work Breakdown Structure and Jira Mapping

1. Identification of WBS Elements

1.1 Main Project (Level 1)

1.0 Upgrade Secure Cloud Data Sharing Platform with Real-Time Breach Detection and Automated Incident Response.

1.2 Major Work Streams (Level 2) and Key Deliverables (Level 3)

WBS Code	Work Stream	Key Deliverables
1.1	Current State Assessment and Requirements Definition	• Security audit report • Risk analysis report • Compliance review report • Vulnerability register • Approved security requirements specification
1.2	Data Protection and Secure Transmission Enhancement	• End-to-end encryption design • Secure API gateway configuration • Key management system implementation • Revised secure data transmission architecture
1.3	Real-Time Breach Detection System	• Intrusion detection mechanism deployment • AI-based anomaly detection module • Centralized log collection pipeline • SIEM integration and alert rules
1.4	Automated Incident Response and Monitoring	• Incident response workflow design • Alerting and escalation rules • Predefined automated response actions • Security reporting and visualization dashboard
1.5	Testing, Validation, and Security Assurance	• Penetration testing report • Vulnerability assessment report • Performance validation report • Defect remediation and sign-off
1.6	Deployment, Training, and Operational Readiness	• Production deployment package • Operational runbook • Administrator and staff training materials

		• User and maintenance documentation
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2. Decomposition

Selected work stream: 1.3 Real-Time Breach Detection System

This work stream is decomposed to Level 4 and Level 5 so that the WBS shows a clear deliverable hierarchy and actual executable work packages.

Level	WBS Code	Element	Type
Level 2	1.3	Real-Time Breach Detection System	Work Stream
Level 3	1.3.1	Intrusion Detection Mechanism Deployment	Deliverable
Level 4	1.3.1.1	Define detection rules and signatures	Sub-deliverable
Level 5	1.3.1.1.1	Configure IDS rule base and thresholds	Work Package
Level 5	1.3.1.1.2	Validate rule coverage against identified threats	Work Package
Level 3	1.3.2	AI-Based Anomaly Detection Module	Deliverable
Level 4	1.3.2.1	Prepare training and validation datasets	Sub-deliverable
Level 5	1.3.2.1.1	Collect and label network and transaction logs	Work Package
Level 5	1.3.2.1.2	Train baseline anomaly detection model	Work Package
Level 5	1.3.2.1.3	Tune detection thresholds for false positives and false negatives	Work Package
Level 3	1.3.3	Centralized Log Monitoring and SIEM Integration	Deliverable
Level 4	1.3.3.1	Build centralized monitoring pipeline	Sub-deliverable
Level 5	1.3.3.1.1	Connect cloud logs, API gateway logs, and security events to SIEM	Work Package
Level 5	1.3.3.1.2	Create dashboards and correlation rules for real-time alerts	Work Package

3. Hierarchical WBS Structure with Codes

1.0 Upgrade Secure Cloud Data Sharing Platform with Real-Time Breach Detection and Automated Incident Response

1.1 Current State Assessment and Requirements Definition

1.1.1 Security audit

1.1.2 Risk analysis

1.1.3 Compliance review

1.1.4 Security requirements specification

1.2 Data Protection and Secure Transmission Enhancement

1.2.1 End-to-end encryption protocols

1.2.2 Secure API gateways

1.2.3 Key management system

1.2.4 Updated transmission architecture

1.3 Real-Time Breach Detection System

1.3.1 Intrusion detection mechanism deployment

1.3.1.1 Define detection rules and signatures

1.3.1.1.1 Configure IDS rule base and thresholds

1.3.1.1.2 Validate rule coverage against identified threats

1.3.2 AI-based anomaly detection module

1.3.2.1 Prepare training and validation datasets

1.3.2.1.1 Collect and label relevant logs

1.3.2.1.2 Train baseline anomaly detection model

1.3.2.1.3 Tune thresholds and evaluate model behavior

1.3.3 Centralized log monitoring and SIEM integration

1.3.3.1 Build centralized monitoring pipeline

1.3.3.1.1 Integrate logs into SIEM

1.3.3.1.2 Build alert dashboards and correlation rules

1.4 Automated Incident Response and Monitoring

1.4.1 Incident response workflow design

1.4.2 Alerting and escalation rules

1.4.3 Automated response actions

1.4.4 Reporting dashboard

1.5 Testing, Validation, and Security Assurance

1.5.1 Penetration testing

1.5.2 Vulnerability assessment

1.5.3 Performance validation

1.5.4 Defect remediation and sign-off

1.6 Deployment, Training, and Operational Readiness

1.6.1 Production deployment

1.6.2 Staff training sessions

1.6.3 User and maintenance documentation

1.6.4 Hypercare and transition to operations

4. WBS Dictionary

Two WBS elements are defined below in a complete and professional dictionary format.

1.2.3 - Key Management System

Description	Design and implement a secure mechanism for generating, storing, rotating, and revoking encryption keys used by the upgraded cloud platform.
Inputs	Approved security requirements, encryption architecture, compliance constraints, existing key handling procedures.
Outputs	Configured key management solution, key lifecycle policy, access controls, implementation records.
Responsible Team	Security Architect, Cloud Security Engineer, Compliance Officer.
Estimated Duration	2 weeks

1.5.1 - Penetration Testing

Description	Execute controlled security attacks on the upgraded solution to validate the effectiveness of encryption, gateway protection, detection controls, and response mechanisms.
Inputs	Deployed test environment, scope and rules of engagement, architecture documents, credentials for authorized testing.
Outputs	Penetration test report, findings log, severity ratings, remediation recommendations.
Responsible Team	Penetration Tester, Security Engineer, QA Lead.
Estimated Duration	1 week

5. Planning Packages vs Work Packages

The case contains both immediately executable work packages and future-oriented planning packages. The distinction below is based on certainty, near-term executability, and maturity of scope.

Item	WBS Reference	Classification	Justification
Configure IDS rule base and thresholds	1.3.1.1.1	Work Package	This is a clearly defined, near-term task with specific inputs, a known team, and a measurable output. It can be scheduled, assigned, and executed immediately.
Penetration testing	1.5.1	Work Package	This activity has a fixed scope before deployment and produces a concrete deliverable, the penetration testing report. Its boundaries and completion criteria are explicit.
Optimize AI detection models	Future enhancement under 1.3.2	Planning Package	The case explicitly states that model optimization will evolve over time as the system matures. Detailed activities are not yet fully decomposed, so it should remain at planning package level.
Integrate external threat intelligence sources	Future enhancement under 1.4 / 1.3	Planning Package	This work is acknowledged as important, but its sources, interfaces, and exact implementation approach are still uncertain. It is therefore a future-oriented placeholder rather than an executable package.

6. Jira Implementation Plan

Selected work stream for Jira representation: 1.3 Real-Time Breach Detection System

6.1 Mapping of WBS to Jira

WBS Element	Mapped Jira Component
Level 2 Work Stream	Epic
Level 3 Deliverable	User Story

Level 4 / Level 5 Detailed Activities	Tasks / Subtasks
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6.2 Epic

Epic Title: Real-Time Breach Detection System

Epic Description: Implement the platform capabilities required to detect suspicious activity in real time through IDS, AI-based anomaly detection, centralized monitoring, and SIEM-driven alerting.

6.3 User Stories with Fields

User Story	Description	Acceptance Criteria	Priority	Assignee	Status
US-1: Deploy intrusion detection coverage	As a security team, we need intrusion detection controls so that malicious or suspicious traffic can be detected during data transmission.	Acceptance Criteria: IDS rules are configured, tested against known attack scenarios, and alerts are generated for defined patterns.	High	Security Engineer	In Progress
US-2: Build AI-based anomaly detection	As a security analyst, I need anomaly detection models so that unusual behavior not covered by static rules can still be identified.	Acceptance Criteria: Baseline model is trained on prepared logs, threshold is defined, and anomalies are visible in monitoring output.	High	AI/ML Engineer	To Do
US-3: Centralize cloud and API security logs	As a SOC team, we need logs centralized in one place so that cross-system events can be correlated in real time.	Acceptance Criteria: Cloud logs, gateway logs, and security events are ingested into the SIEM with normalized timestamps and searchable fields.	High	DevOps Engineer	To Do
US-4: Configure SIEM alert rules and dashboards	As a security operations team, we need real-time alerts	Acceptance Criteria: Correlation rules are active,	Medium	SOC Analyst	To Do

	and dashboards so that breach indicators are visible and actionable.	critical alerts reach the correct team, and dashboard widgets show live security status.			
US-5: Validate detection effectiveness	As the project team, we need validation of detection performance so that the deployed solution is reliable before production release.	Acceptance Criteria: Detection tests are executed, false positives are reviewed, and findings are documented for remediation.	Medium	QA Lead	To Do

6.4 Example Tasks / Subtasks under the Epic

- Task: Create IDS rule inventory, Subtasks: draft rules, review rules, deploy to test environment.
- Task: Prepare anomaly detection dataset, Subtasks: extract logs, label anomalies, split validation data.
- Task: Integrate SIEM data sources, Subtasks: connect API gateway logs, connect application logs, verify field mapping.
- Task: Build dashboard, Subtasks: create alert widgets, create trend graphs, validate live refresh.

6.5 Screenshot Placeholders.

Screenshot 1: Epic and backlog structure

The screenshot displays the Jira interface for an Epic titled "Real-Time Breach Detection System Implementation" within the "Cloud Security Class A..." space. The interface includes a sidebar with navigation options like "For you", "Spaces", and "Recent". The main content area shows the Epic's details, including its due date (Mar 28, 2026), description, objective, context, and scope. The scope is divided into "Included" and "Not Included" sections. The right sidebar contains sections for "Details", "Development", and "Automation", with a "Recent rule runs" section at the bottom.

Epic Details:

- Title:** Real-Time Breach Detection System Implementation
- Due date:** Mar 28, 2026
- Description:** Implement a comprehensive breach detection system.
- Context:** The project involves integrating various technologies for effective breach detection. This includes Intrusion Detection Systems (IDS), AI-based anomaly detection, and Security Information and Event Management (SIEM) monitoring.
- Scope:**
 - Included:**
 - Implementation of IDS
 - Development of AI-based anomaly detection
 - Integration of SIEM monitoring
 - Not Included:**

Right Sidebar:

- Details:**
 - Priority: High
 - Assignee: Muhammad Zayyam Hass...
 - Labels: None
 - Reporter: Muhammad Zayyam Hass...
- Development:**
 - Connect development tools
- Automation:**
 - Recent rule runs
 - Refresh to see recent runs.
 - Create new automation rule

Screenshot 2: User Stories with fields

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zayyamhassan-1774508592934.atlassian.net/browse/SCRUM-8

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Surah Ya seen Sum... C++ Programming... Pointer to an Array [...]

Jira

For you Spaces + ... Recent Cloud Security Class A... More spaces > Recommended Collect requests TRY ... More

Cloud Security Class... / SCRUM-6 / SCRUM-8

Deploy Intrusion Detection System

+ ...

Description

User story

- As a security engineer, when configuring the IDS, I want to deploy it so that we can monitor network traffic for suspicious activities.

Context

The issue involves the configuration and deployment of an Intrusion Detection System (IDS) to enhance network security.

Acceptance criteria

- IDS is fully configured with custom rules.
- IDS successfully detects test intrusion attempts.

Other information

N/A

Subtasks

Create suggested work items View 4 suggestions

0% Done

Work	Priority	Assignee	Status
SCRUM-9 Install IDS software	Medium	Unassigned	To Do
SCRUM-10 Configure detection rules and policies	Medium	Unassigned	To Do
SCRUM-11 Test IDS with simulated attacks	Medium	Unassigned	To Do

What needs to be done? Subtask Cancel

Choose existing

Linked work items

To Do ⚙️ Improve Story

Details

Assignee Muhammad Zayyam Hassan

Priority Medium

Parent SCRUM-6 Real-Time Breach Detection System

Due date None

Labels None

Team None

Start date None

Sprint None

Story point estimate None

Reporter Muhammad Zayyam Hassan

Development

Connect development tools

Automation

Recent rule runs Refresh to see recent runs. Create new automation rule

Created 1 minute ago Updated 1 minute ago Configure

← → ↺

zayyamhassan-1774508592934.atlassian.net/browse/SCRUM-12

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Jira

For you Spaces + ... Recent Cloud Security Class A... More spaces > Recommended Collect requests TRY ... More

Cloud Security Class... / SCRUM-6 / SCRUM-12

Develop AI Anomaly Detection Model

+ ...

Description

User story

- As a data scientist, when building an AI model, I want to detect anomalies in network behavior automatically so that I can improve network security.

Context

The goal is to develop an AI anomaly detection model that can identify unusual patterns in network behavior. This model will help enhance the security and reliability of the network.

Acceptance criteria

- Model achieves a minimum of 95% accuracy on the test dataset.
- Model integrates with the detection system.

Other information

N/A

Subtasks

Create suggested work items View 6 suggestions

0% Done

Work	Priority	Assignee	Status
SCRUM-13 Collect and prepare training data	Medium	Unassigned	To Do
SCRUM-14 Train machine learning model	Medium	Unassigned	To Do
SCRUM-15 Validate model performance	Medium	Unassigned	To Do
SCRUM-16 Deploy model to production environment	Medium	Unassigned	To Do

What needs to be done? Subtask

To Do ⚙️ Improve Story

Details

Assignee Unassigned Assign to me

Priority High

Parent SCRUM-6 Real-Time Breach Detection System

Due date None

Labels None

Team None

Start date None

Sprint None

Story point estimate None

Reporter Muhammad Zayyam Hassan

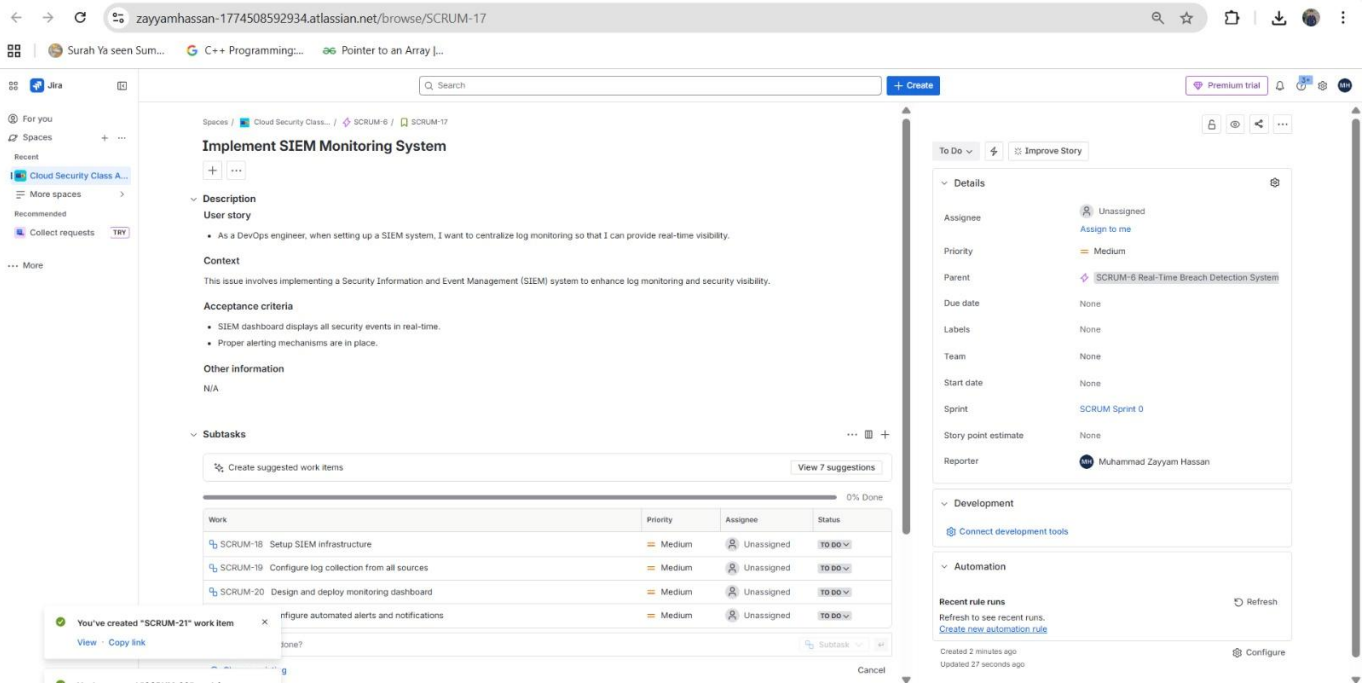
Development

Connect development tools

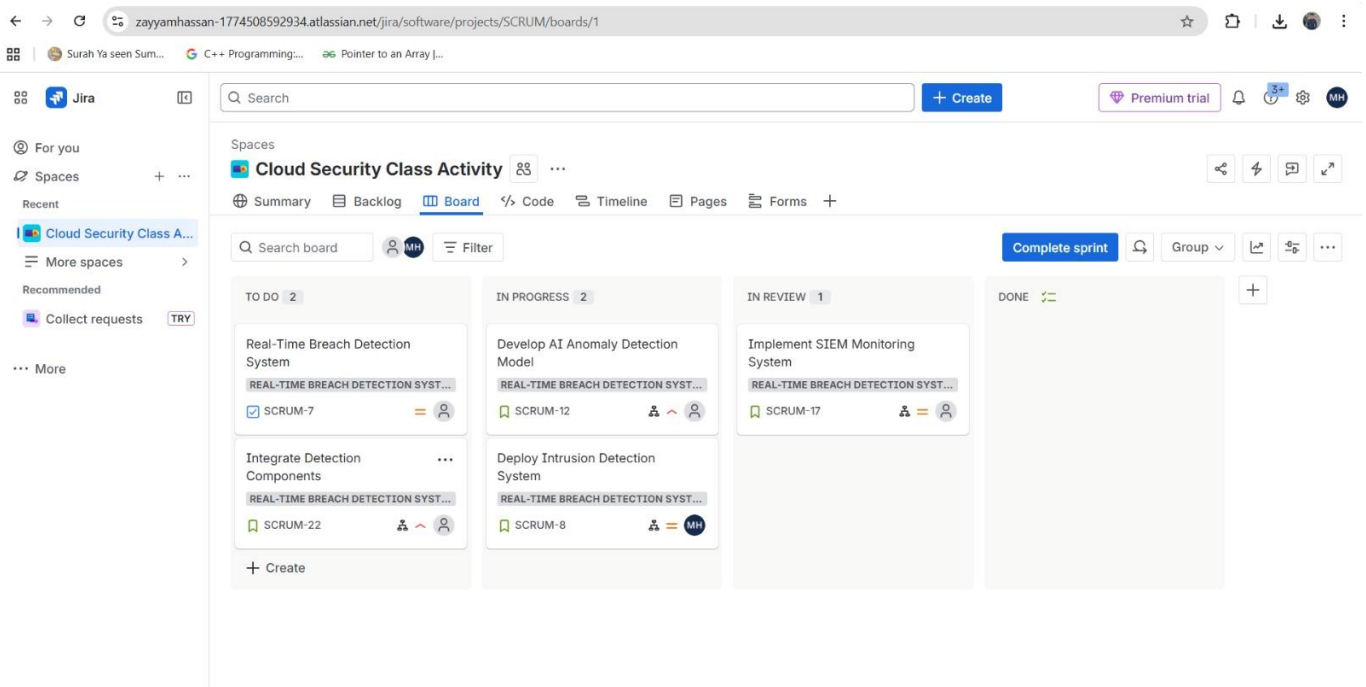
Automation

Recent rule runs Refresh to see recent runs. Create new automation rule

Created 8 seconds ago Updated 8 seconds ago Configure



Screenshot 3: Tasks / Subtasks view



7. Final Deliverables Summary

The final project deliverables are fully aligned with the case: upgraded secure cloud architecture, end-to-end encryption, secure API gateway, key management system, real-time breach detection capability, centralized SIEM monitoring, automated incident response workflows, reporting dashboard, validated test results, production deployment, training, and operational documentation.

8. Critical Thinking Note

A strong WBS should include only defined deliverables and executable work, while still acknowledging uncertain future scope through planning packages. In this case, the core implementation work is sufficiently mature to decompose into deliverables and work packages, but AI optimization, threat intelligence integration, and scalability should remain controlled placeholders until operational evidence and technical decisions become clearer. This distinction improves planning accuracy and keeps the WBS realistic and manageable.